

EMBERGUARD · DESIGN DECISIONS REGISTER

EmberGuard design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Approved follow-ups carried out; value engineering section updated to USD 707 with the sun shade and the 45 °C charge controller priced
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 12 (EGD-DDR-003 accepted; pitch reworded); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

EmberGuard design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The thermal sensor breakout is no larger than 25 x 25 mm, with the sensor can centred and four corner holes	It mounts on the inside of the pod's west wall with its can in a 9.5 mm window	EGD-DDR-003, P6
2	The die-cast pod box has walls of 4 mm or more and its lid screws into corner bosses	The sensor and lens hood screw into tapped holes in the wall; the hood spacers into the lid	EGD-DDR-003, P6, P7
3	The slip-on base flange fits 33.7 mm pipe, its four holes lie on an 80 mm circle and it has set screws	The wall plate's flange holes and the standoff fit depend on it	EGD-DDR-003, P1
4	The U-bolts' leg spacing for 40 mm tube and for 33.7 mm pipe	They set the eight holes in each crossover plate	EGD-DDR-003, P1
5	The steel enclosure comes with a gear plate on studs and a four-lug wall kit	The battery strap, board standoffs and wall fixing use them	EGD-DDR-003, P9
6	The solar panel frame accepts the pole-mount tilt bracket	The panel's only fixing	EGD-DDR-003, P3
7	The zone valves have inlet and outlet in line, top and bottom, with the coil to one side	The valves hang from the manifold tees	EGD-DDR-003, P10
8	The gutter-lip clips fit the gutter's lip profile	They carry the eave line	EGD-DDR-003, P11
9	At the house: a sound timber verge board at each gutter end, a fascia lower edge for the riser clip, and a gable wall that takes the wall anchors (about 535 N pull each)	The pods, risers and mast fix to them	EGD-DDR-003, P1, P5, P11; EGD-CAL-001, E4
10	The charge controller takes a battery temperature probe and stops charging above 45 °C	The enclosure reaches about 63 °C in sun at 60 °C ambient, above the pack's charge limit	BOM line 9; EGD-CAL-001, F6
11	The gable wall takes two M6 plugs for the sun shade (about 320 N pull each at 120 km/h)	The shade's only fixing	BOM line 18; EGD-CAL-001, E7

■ Value engineering

Value-engineering target: USD 605. Estimated cost of the constructable design: USD 707 (USD 102 over the target). The target is a hypothetical control target, not a limit. Main cost drivers and savings worth trying:

- Of the USD 102 over the target, USD 84 is the parts added to make the kit buildable (EGD-DDR-003): flanges, crossover plates and U-bolts (+USD 32), pod mounts and lens hoods (+USD 18), the panel tilt bracket (+USD 6), longer cables (+USD 3), and the valve board, clips, glands and strap (+USD 25).
- The largest lines are the two thermal sensors (USD 96), the mast, standoffs and crossover plates (USD 80), the two sensor pods (USD 74) and the spray lines and heads (USD 52).
- Decided on 2026-10-02 and now priced: the folded white shade for the ground enclosure (line 18, USD 14: sheet USD 5, flat bar USD 3, paint USD 2, screws, plugs and rivets USD 4) and a charge controller with a battery temperature probe that stops charging above 45 °C (line 9, USD 4 more).
- Savings worth trying, about USD 35 in all: bolt the flanges straight to the wall without wall plates (about USD 10, but the anchor pull roughly doubles and must be rechecked), a plywood valve board (about USD 12), and unbranded thermal breakouts (about USD 10 to USD 15).
- Metal eave runs would add USD 110 and are a priced option outside the kit, not a saving.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D9: budget \$425, two MLX90640 thermal arrays, gable-end mast, metal eave runs if the budget allows (it did not), tank with its own pump, pump power out of scope, adjustable arming defaults, fail to wet on sensor loss, normally closed valves	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	EGD-DDR-001
2026-09-25	O2 to O5: budget \$575, thermal sensors in gutter-corner pods, leeward zone only in wind, 2 K trigger and 10 Ah battery	Amish: "i accept all your recommendations, go with them across all repos."	EGD-DDR-002
2026-09-26	Budget set to \$605 to cover the priced BOM (N1)	Amish: "i approve all the budget items."	EGD-DDR-002
2026-09-26	EmberGuard chosen for the first batch of product renders	Amish	Review note, 2026-09-26
2026-10-02	Design for construction accepted: the changes P1 to P12, as made, including P10, which feeds the two zone valves side by side	Amish: "i approve your recommendations for all 555 open decisions."	EGD-DDR-003, Table 1 (P1 to P12)
2026-10-02	No window in front of the sensor for the prototype; an ash-fouled lens case is included in the TRL 4 heated-target and ash trials, and the window is decided from that result	Amish: "i approve your recommendations for all 555 open decisions."	EGD-DDR-003, A2
2026-10-02	Pod mount: the verge cleat now, with a sound timber verge confirmed at each site; a fascia-face bracket is drawn once the partner's houses are known	Amish: "i approve your recommendations for all 555 open decisions."	EGD-DDR-003, A3
2026-10-02	Ground enclosure: a light finish and the shadiest available wall now; the folded white shade is designed at the next revision; the charge controller must stop charging above 45 °C	Amish: "i approve your recommendations for all 555 open decisions."	EGD-CAL-001, F1; review note, 2026-09-25
2026-10-02	Leeward wetting: larger, lower-angle droplets and a second leeward row are studied on paper at the next revision; R6 stays at the 30 km/h design wind	Amish: "i approve your recommendations for all 555 open decisions."	EGD-DDR-002, N2
2026-10-02	Deep gutter debris: raising the pods to about 1 m above the gutter lip is studied on paper (including the arm's wind load and ember exposure); gutter guards are listed in the installation notes now	Amish: "i approve your recommendations for all 555 open decisions."	EGD-DDR-002, N3
2026-10-02	Pitch adopted: "Gutter-corner thermal sensors and a weather mast that detect ember showers and trigger gutter and eave sprinkler zones."	Amish: "i approve your recommendations for all 555 open decisions."	EGD-DDR-002, N4
2026-10-02	First co-design partner to approach: a recognised Firewise USA neighbourhood group in a wildland-urban interface area whose houses have gable ends with timber verges; in Texas, through the Texas A&M Forest Service, which supports such groups	Amish: "i approve your recommendations for all 555 open decisions."	EGD-DDR-001, O1

Date	Decision	Decided by	Record
2026-10-02	Product render layout accepted as render-only, with captions saying the mast and ground unit are drawn closer than installed	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 1
2026-10-02	Spray lines stay blue in the concept media; the BOM still specifies black line	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 5
2026-10-02	A third sensor for vents, decks and the far gable is left to a later version	Amish: "i approve your recommendations for all 555 open decisions."	EGD-PRC-001, open questions
2026-10-02	Ember arrival data sharing is opt-in, with the first partner agreeing what is shared and with whom	Amish: "i approve your recommendations for all 555 open decisions."	EGD-PRC-001, open questions